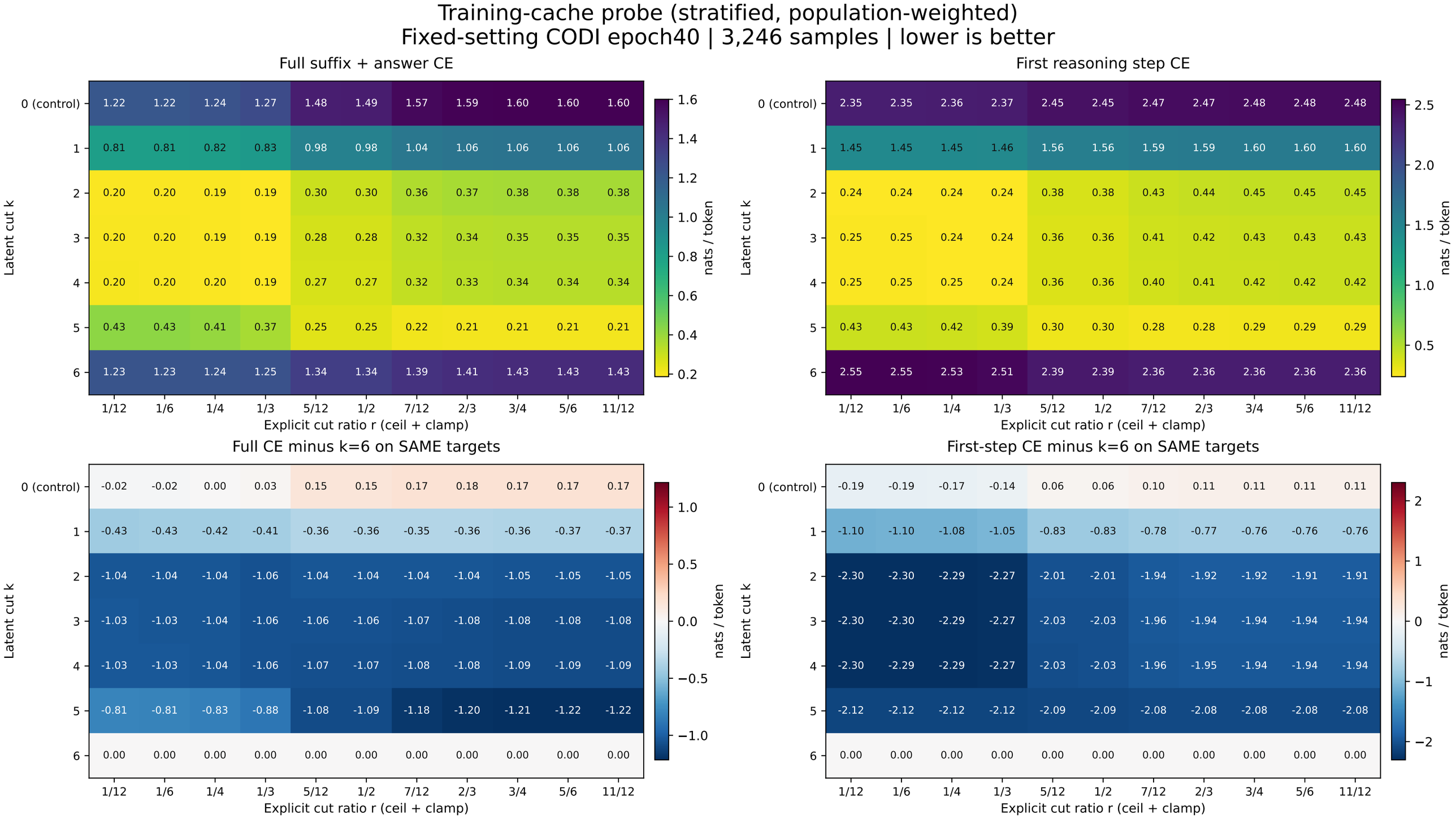
Training-cache probe (stratified, population-weighted)
Fixed-setting CODI epoch40 | 3,246 samples | lower is better
Full suffix + answer CE
0 (control) 1.22 1.22 1.24 1.27 1.48 1.49 1.57 1.59 1.60 1.60 1.60
1 0.81 0.81 0.82 0.83 0.98 0.98 1.04 1.06 1.06 1.06 1.06
2 0.20 0.20 0.19 0.19 0.30 0.30 0.36 0.37 0.38 0.38 0.38
3 0.20 0.20 0.19 0.19 0.28 0.28 0.32 0.34 0.35 0.35 0.35
4 0.20 0.20 0.20 0.19 0.27 0.27 0.32 0.33 0.34 0.34 0.34
5 0.43 0.43 0.41 0.37 0.25 0.25 0.22 0.21 0.21 0.21 0.21
6 1.23 1.23 1.24 1.25 1.34 1.34 1.39 1.41 1.43 1.43 1.43
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
1.6 1.4 1.2 1.0 0.8 0.6 0.4 0.2
nats / token
First reasoning step CE
0 (control) 2.35 2.35 2.36 2.37 2.45 2.45 2.47 2.47 2.48 2.48 2.48
1 1.45 1.45 1.45 1.46 1.56 1.56 1.59 1.59 1.60 1.60 1.60
2 0.24 0.24 0.24 0.24 0.38 0.38 0.43 0.44 0.45 0.45 0.45
3 0.25 0.25 0.24 0.24 0.36 0.36 0.41 0.42 0.43 0.43 0.43
4 0.25 0.25 0.25 0.24 0.36 0.36 0.40 0.41 0.42 0.42 0.42
5 0.43 0.43 0.42 0.39 0.30 0.30 0.28 0.28 0.29 0.29 0.29
6 2.55 2.55 2.53 2.51 2.39 2.39 2.36 2.36 2.36 2.36 2.36
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
2.5 2.0 1.5 1.0 0.5
nats / token
Full CE minus k=6 on SAME targets
0 (control) -0.02 -0.02 0.00 0.03 0.15 0.15 0.17 0.18 0.17 0.17 0.17
1 -0.43 -0.43 -0.42 -0.41 -0.36 -0.36 -0.35 -0.36 -0.36 -0.37 -0.37
2 -1.04 -1.04 -1.04 -1.06 -1.04 -1.04 -1.04 -1.04 -1.05 -1.05 -1.05
3 -1.03 -1.03 -1.04 -1.06 -1.06 -1.06 -1.07 -1.08 -1.08 -1.08 -1.08
4 -1.03 -1.03 -1.04 -1.06 -1.07 -1.07 -1.08 -1.08 -1.09 -1.09 -1.09
5 -0.81 -0.81 -0.83 -0.88 -1.08 -1.09 -1.18 -1.20 -1.21 -1.22 -1.22
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
1.0 0.5 0.0 −0.5 −1.0
nats / token
First-step CE minus k=6 on SAME targets
0 (control) -0.19 -0.19 -0.17 -0.14 0.06 0.06 0.10 0.11 0.11 0.11 0.11
1 -1.10 -1.10 -1.08 -1.05 -0.83 -0.83 -0.78 -0.77 -0.76 -0.76 -0.76
2 -2.30 -2.30 -2.29 -2.27 -2.01 -2.01 -1.94 -1.92 -1.92 -1.91 -1.91
3 -2.30 -2.30 -2.29 -2.27 -2.03 -2.03 -1.96 -1.94 -1.94 -1.94 -1.94
4 -2.30 -2.29 -2.29 -2.27 -2.03 -2.03 -1.96 -1.95 -1.94 -1.94 -1.94
5 -2.12 -2.12 -2.12 -2.12 -2.09 -2.09 -2.08 -2.08 -2.08 -2.08 -2.08
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
2 1 0 −1 −2
nats / token